United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Final-Term Exam: Trimester: Fall 2025

Course Code: EEE 2123; **Course Title:** Electronics
Total Marks: 40; **Duration:** 2hr

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

There are six (6) questions in this paper. Answer all of them.

1. Consider an NMOS transistor fabricated in a 0.18-µm process with L = 0.18 µm and W = 2 µm. The process technology is specified to have C_{ox}= 8.6 fF/µm^2, μ_n = 450 cm^2/V.s and V_{tn} =0.5 V.
   (a) Find V_{GS} and V_{DS} that result in the MOSFET operating at the edge of saturation with I_D=100 µA. **[4]**
   (b) If V_{GS} is kept constant, find V_{DS} that results in I_D= 50µA. (Triode region) **[3]**

2. From the circuit of Fig 2,
   a. Analyze the circuit (Fig 2) to determine I_D, I_S, V_D, V_S, V_G, and I_G. Let V_t = 1.2V, $K'_n(W/L) = 1\ mA/V^2$. Assume λ = 0. **[5]**
   b. What is the minimum value of R_D required for the MOSFET to operate in the linear region? **[5]**

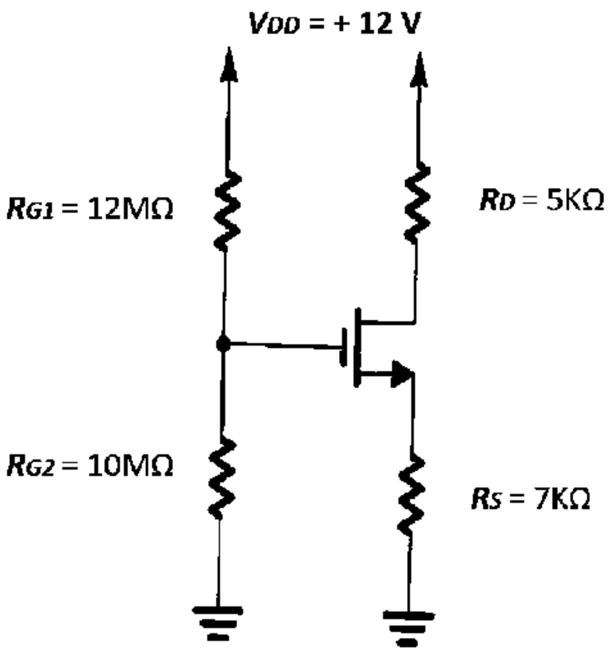

Fig 2: Figure for question 2

3. From the circuit of Fig 3, Find all node voltages and branch currents in the following circuit, $\beta = 100$ **[6]**

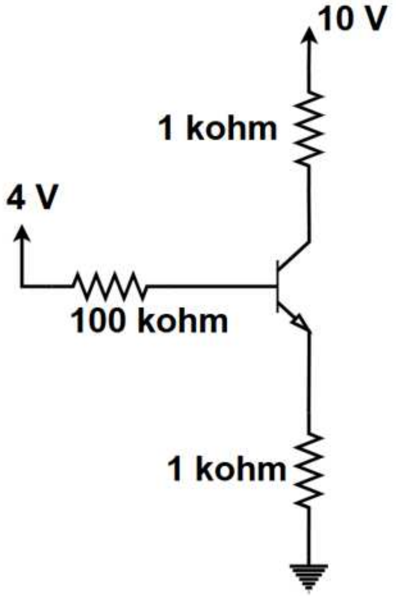

Fig 3: Figure for question 3

4. From the circuit of Fig 4, The BJT in Fig 4 is deep in saturation. Find the values of α and β of the BJT. **[5]**

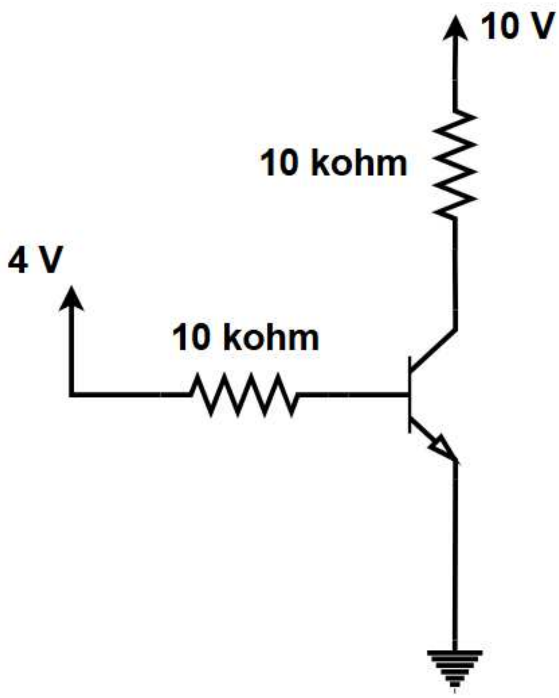

Fig 4: Figure for question 4

5. (a) From the circuit of Fig 5, The **PMOS** network of a random digital system is given below. Draw the **NMOS** network and determine the output logic. **[3]**

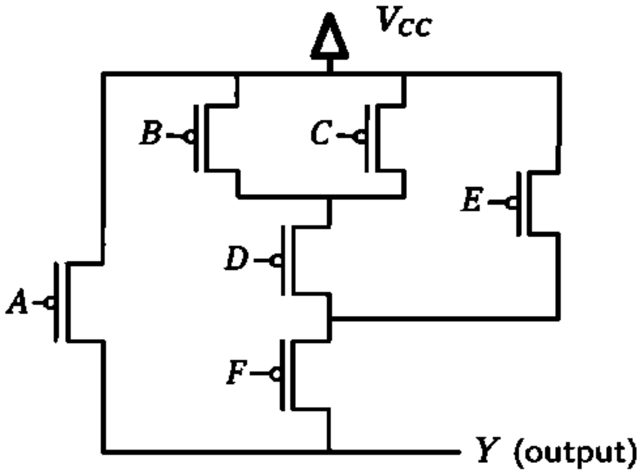

Fig 5: Figure for question 5 (a)

(b) Implement the following logic function using CMOS: **[3]**

$$Y = ((a + b)\,\overline{c} + de)\,\overline{f}$$

6. Implement the following equation using Op-Amps: **[6]**

$$V_o = 2V_1 - 3V_2$$